Week 1

- **Problem 1: Conjunction introduction.**
  Let $a, b \in \mathbb{R}$. Suppose that

  $$a + b = 8, \quad ab = 12.$$

  Prove that

  $$a > 0, \quad b > 0 \quad \text{and} \quad |a - b| = 4.$$

  - **Solution 1.1**
    Since a and b satisfy $a + b = 8$ and $ab = 12$, according to Vieta's formulas, they are the roots of the quadratic equation:

    $$t^2 - 8t + 12 = 0$$

    Factoring the equation gives:

    $$(t - 2)(t - 6) = 0$$

    Thus, the roots are 2 and 6. This implies that either $\{a, b\} = \{2, 6\}$.

    In both cases, we can check each part of the conjunction:

    1. $a > 0$ and $b > 0$ since both $2 > 0$ and $6 > 0$ are true.
    2. $|a - b| = |2 - 6| = |-4| = 4$ (or $|6 - 2| = 4$).

    By conjunction introduction, since all components are true, the statement holds:

    $$a > 0 \land b > 0 \land |a - b| = 4$$

- **Problem 2: Disjunction introduction.**
  Consider the equation

  $$x^3 - ax = 0,$$

  where $a \in \mathbb{R}$.
  Prove that, for every value of a, the equation has either exactly one or exactly three distinct real solutions. Determine for which values of a each case occurs.

  - **Solution 2.1**
    We can factor the given equation as:

    $$x(x^2 - a) = 0$$

    This gives one guaranteed real solution:

    $$x = 0$$

    The remaining solutions depend on the quadratic part:

    $$x^2 = a$$

    We separate this into cases based on the value of a:

    - Case 1: If $a \leqslant 0$, then $x^2 = a$ has no distinct real solutions other than $x = 0$ (when $a = 0$, it is a repeated root). Thus, there is exactly one real solution.
    - Case 2: If $a > 0$, then $x^2 = a$ yields two distinct real solutions: $x = \sqrt{a}$ and $x = -\sqrt{a}$.

Since $a \neq 0$, these are distinct from $x = 0$. Thus, there are exactly three distinct real solutions.

Therefore, for any $a \in \mathbb{R}$, the equation has either exactly one solution (when $a \leqslant 0$) or exactly three solutions (when $a > 0$).

- **Problem 3: Disjunction elimination: proof by cases.**
  Let $a_1, a_2, \dots, a_5 \in \mathbb{Z}$. Prove that among these five integers, one can always choose four numbers whose sum is even.
  - **Solution 3.1**
    Let K be the number of odd integers among the five given integers $a_1, a_2, \dots, a_5$.
    The number of even integers is $5 - K$.
    The possible values for K range from 0 to 5. We analyze the problem by partitioning K into the following three cases:
    - **Case 1:** $K \leqslant 1$.
      The number of even integers is $5 - K \geqslant 4$. We can choose 4 even numbers, which can be written as $2n_1, 2n_2, 2n_3, 2n_4$ for some $n_1, n_2, n_3, n_4 \in \mathbb{Z}$. Their sum is:
      $$2n_1 + 2n_2 + 2n_3 + 2n_4 = 2(n_1 + n_2 + n_3 + n_4)$$
      Since $n_1 + n_2 + n_3 + n_4 \in \mathbb{Z}$, the sum is even.
    - **Case 2:** $K \geqslant 4$.
      There are at least 4 odd numbers available. We can choose 4 odd numbers, which can be written as $2n_1 + 1, 2n_2 + 1, 2n_3 + 1, 2n_4 + 1$ for some $n_1, n_2, n_3, n_4 \in \mathbb{Z}$. Their sum is:
      $$(2n_1 + 1) + (2n_2 + 1) + (2n_3 + 1) + (2n_4 + 1) = 2(n_1 + n_2 + n_3 + n_4 + 2)$$
      Since $n_1 + n_2 + n_3 + n_4 + 2 \in \mathbb{Z}$, the sum is even.
    - **Case 3:** $2 \leqslant K \leqslant 3$.
      There are at least 2 odd numbers and at least 2 even numbers available. We can choose 2 even numbers ($2n_1, 2n_2$) and 2 odd numbers ($2n_3 + 1, 2n_4 + 1$) for some n_1, n_2, n_3, n_4 $\in \mathbb{Z}$. Their sum is:
      $$2n_1 + 2n_2 + (2n_3 + 1) + (2n_4 + 1) = 2(n_1 + n_2 + n_3 + n_4 + 1)$$
      Since $n_1 + n_2 + n_3 + n_4 + 1 \in \mathbb{Z}$, the sum is even.

    Since all possible values of K fall into one of these three cases, and each case mathematically guarantees an even sum, the proof is complete.

- **Problem 4**
  Construct a truth table and prove that
  $$(P \to Q) \land (P \to R) \equiv P \to (Q \land R).$$
  - **Solution 4.1**
    The truth table for both expressions is constructed below:

P	Q	R	$P \to Q$	$P \to R$	$(P \to Q) \land (P \to R)$	$Q \land R$	$P \to (Q \land R)$
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	0
1	0	1	0	1	0	0	0
1	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1
0	1	0	1	1	1	0	1
0	0	1	1	1	1	0	1
0	0	0	1	1	1	0	1

Since the columns for $(P \to Q) \land (P \to R)$ and $P \to (Q \land R)$ are identical in every row, the two statements are logically equivalent.

- **Problem 5**
  Let the universe of discourse be

$$X = \{a, b\},$$

  and let $P(x)$ and $Q(x)$ be mathematical statements depending on $x \in X$.
  Using a truth table, determine whether the statements

$$\forall x \in X\,(P(x) \land Q(x))$$

  and

$$(\forall x \in X\,P(x)) \land (\forall x \in X\,Q(x))$$

  are logically equivalent.
  First rewrite each quantified statement using only $P(a), P(b), Q(a)$, and $Q(b)$, and then construct the truth table.

  - **Solution 5.1**
    First, we expand the quantified statements over the finite domain $X = \{a, b\}$:
    1. $\forall x \in X\,(P(x) \land Q(x)) \equiv (P(a) \land Q(a)) \land (P(b) \land Q(b))$
    2. $(\forall x \in X\,P(x)) \land (\forall x \in X\,Q(x)) \equiv (P(a) \land P(b)) \land (Q(a) \land Q(b))$

    By the associative and commutative laws of conjunction ($\land$), both expressions simplify directly to:

$$P(a) \land P(b) \land Q(a) \land Q(b)$$

    To keep the table compact and readable, let

$$\begin{array}{lll} A \equiv P(a) \land Q(a) & B \equiv P(b) \land Q(b) & C \equiv \forall x \in X\,(P(x) \land Q(x)) \\ D \equiv \forall x \in X\,P(x) & E \equiv \forall x \in X\,Q(x) & F \equiv (\forall x \in X\,P(x)) \land (\forall x \in X\,Q(x)) \end{array}$$

    Thus, they are logically equivalent. The corresponding truth table configuration for their components is:

$P(a)$	$P(b)$	$Q(a)$	$Q(b)$	A	B	C	D	E	F
1	1	1	1	1	1	1	1	1	1
1	1	1	0	1	0	0	1	0	0
1	1	0	1	0	1	0	1	0	0
1	1	0	0	0	0	0	1	0	0
1	0	1	1	1	0	0	0	1	0
1	0	1	0	1	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0
0	1	1	1	0	1	0	0	1	0
0	1	1	0	0	0	0	0	0	0
0	1	0	1	0	1	0	0	0	0
0	1	0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0	1	0
0	0	1	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

Therefore, the statements are logically equivalent.